LAB EXPERIMENT RECORD

Set 24 Product Inventory Management

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
Name	Swetha S
Register Number	192424049
Date	15/10/2026
Day	Thursaday

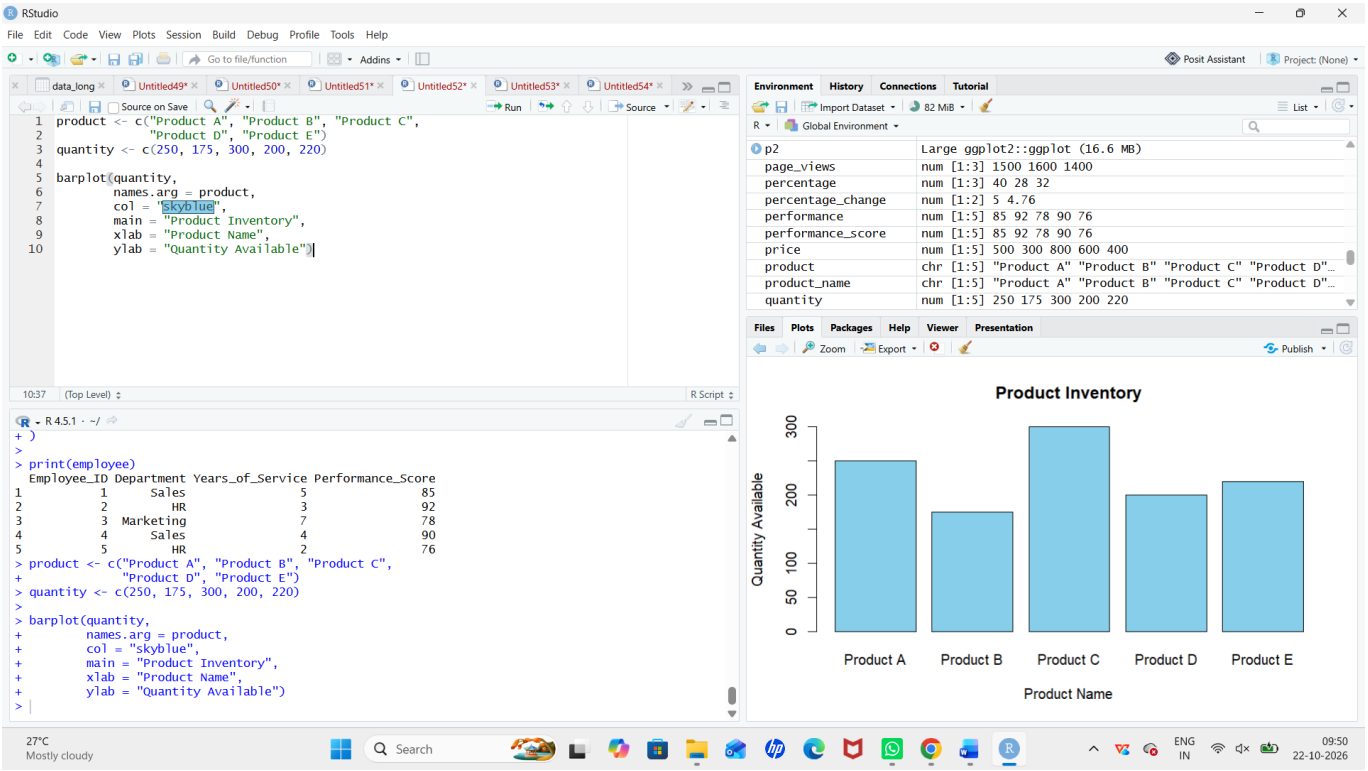
AIM

To analyze product inventory data using R and Tableau by creating a bar chart, a stacked bar chart, a scatter plot, and an interactive dashboard to understand product quantities, category-wise inventory distribution, and the relationship between product price and quantity available.

1.

```
product <- c("Product A", "Product B", "Product C",
             "Product D", "Product E")
quantity <- c(250, 175, 300, 200, 220)

barplot(quantity,
        names.arg = product,
        col = "skyblue",
        main = "Product Inventory",
        xlab = "Product Name",
        ylab = "Quantity Available")
```

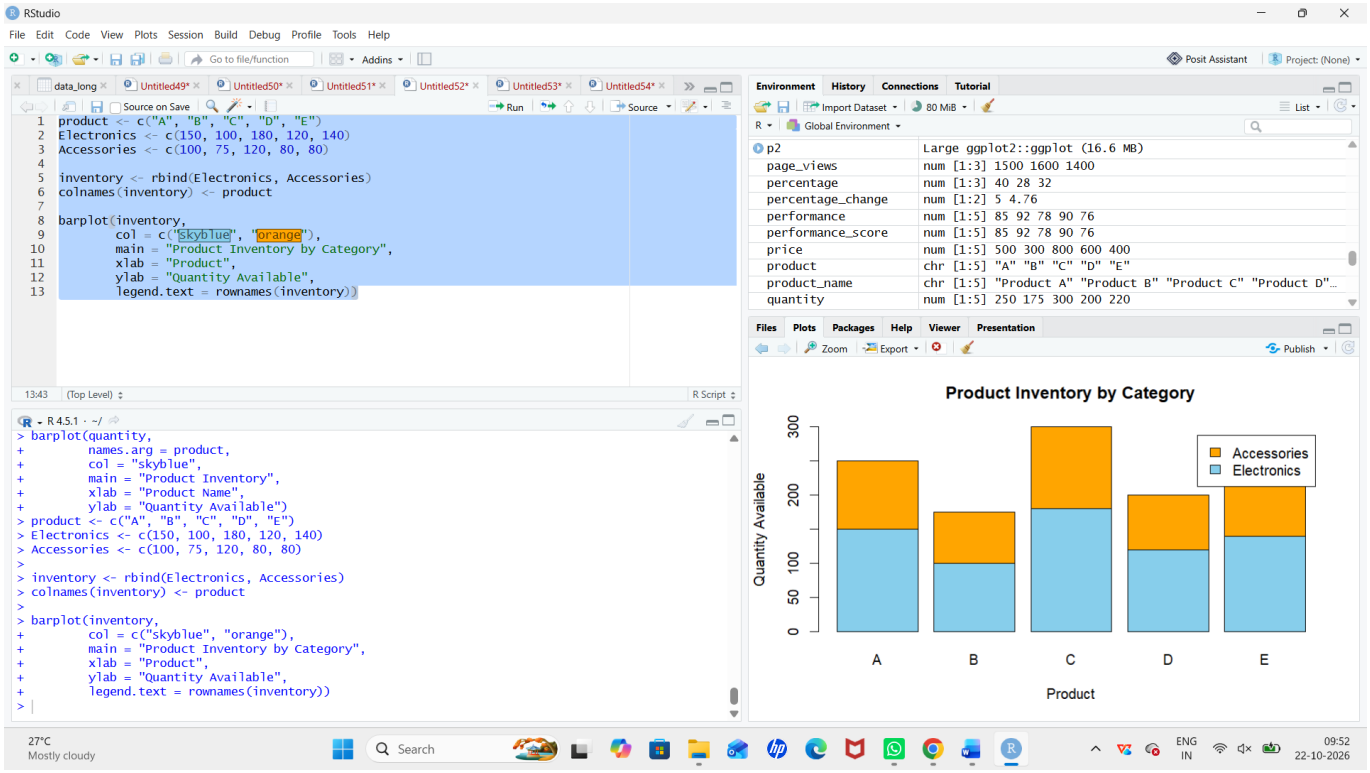


2

```
product <- c("A", "B", "C", "D", "E")
Electronics <- c(150, 100, 180, 120, 140)
Accessories <- c(100, 75, 120, 80, 80)

inventory <- rbind(Electronics, Accessories)
colnames(inventory) <- product

barplot(inventory,
        col = c("skyblue", "orange"),
        main = "Product Inventory by Category",
        xlab = "Product",
        ylab = "Quantity Available",
        legend.text = rownames(inventory))
```



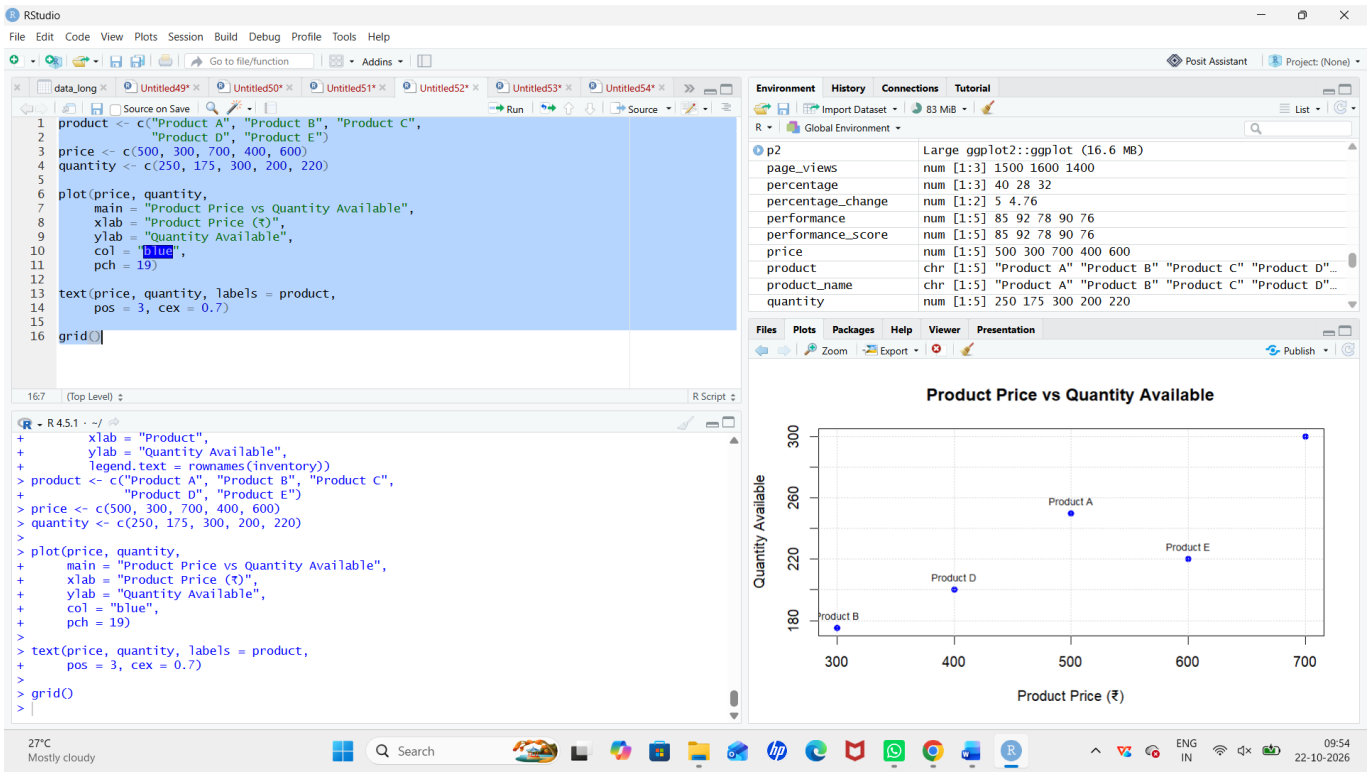
4.

```
product <- c("Product A", "Product B", "Product C",
             "Product D", "Product E")
price <- c(500, 300, 700, 400, 600)
quantity <- c(250, 175, 300, 200, 220)

plot(price, quantity,
     main = "Product Price vs Quantity Available",
     xlab = "Product Price (₹)",
     ylab = "Quantity Available",
     col = "blue",
     pch = 19)

text(price, quantity, labels = product,
     pos = 3, cex = 0.7)

grid()
```



3.

 Create an Excel or CSV file containing Product ID, Product Name, and Quantity Available.

 Add Product Category if you want to create the stacked bar chart.

 Open Tableau and connect to the dataset.

 Create Sheet 1 – Inventory Bar Chart by dragging Product Name to Columns and Quantity Available to Rows.

 Create Sheet 2 – Stacked Bar Chart by dragging Product Name to Columns, Quantity Available to Rows, and Product Category to Color.

 Open a new dashboard and add both worksheets.

 Add suitable titles, labels, colors, and filters for interactive exploration.

 Arrange the charts clearly and save the dashboard.

Procedure

1. Create the product inventory dataset containing Product ID, Product Name, and Quantity Available.

2. Define the dataset in R using a data frame or vectors.

3. Create a bar chart to visualize the quantity available for each product.

4. Generate a stacked bar chart to display product quantities across different product categories.

5. Create a scatter plot to explore the relationship between product price and quantity available using sample price data.

6. Import the inventory dataset into Tableau.

7. Create separate worksheets for the bar chart and stacked bar chart.

8. Combine both charts into a single dashboard with suitable titles, labels, and filters.

9. Analyze the visualizations to compare product stock levels and identify inventory patterns.

Result

Product inventory data was successfully analyzed using R and Tableau. The bar chart displayed the quantity available for each product, the stacked bar chart showed category-wise inventory distribution, and the scatter plot helped explore the relationship between product price and quantity. The Tableau dashboard combined the visualizations for easy inventory monitoring.